WORKED SOLUTION

STEP II 2024, Question 9

Mechanics • Projectile Motion

STEP II 2024, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

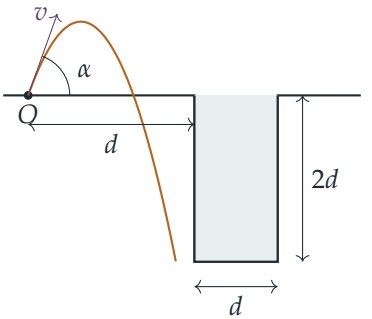
www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything follows from the trajectory equation $y = x\tan\alpha - \dfrac{gx^2}{2v^2}\sec^2\alpha$, written as a quadratic in $t = \tan\alpha$. Landing on the base cleanly means two things: the particle is still above ground level when it reaches the near edge, and it is already below the level of the base by the time it would reach the far side. Each condition is a quadratic inequality in t, and completing the square turns them into the forms we need. Part (ii) is then just a matter of showing the two allowed ranges of t overlap.

Solution

Model the particle as a point projectile under gravity alone, with no air resistance. Take the origin at the point of projection, x horizontally towards the trench and y vertically upwards. The near wall is then the line $x = d$, the far wall $x = 2d$, and the base $y = -2d$ for $d < x < 2d$.



With horizontal velocity $v\cos\alpha$ and vertical velocity $v\sin\alpha - gT$ at time T, eliminating T gives

$$y = x\tan\alpha - \frac{gx^2}{2v^2\cos^2\alpha} = xt - \frac{x^2(1+t^2)}{2\lambda d},$$

where $t = \tan\alpha$ and we have used $v^2 = \lambda dg$ and $\sec^2\alpha = 1 + t^2$.

Part (i)(a)

To avoid the ground before the trench and the near wall, the particle must still be strictly above ground level when $x = d$. (If it were at or below $y = 0$ there, it would already have struck the ground or would strike the near wall.) So

$$y(d) = dt - \frac{d(1+t^2)}{2\lambda} > 0 \iff 2\lambda t - 1 - t^2 > 0 \iff t^2 - 2\lambda t + \lambda^2 < \lambda^2 - 1,$$

multiplying by $2\lambda/d > 0$. That is,

$$(\tan\alpha - \lambda)^2 < \lambda^2 - 1,$$

as required. The left-hand side is a square, so $\lambda^2 - 1 > 0$; since $\lambda = v^2/(dg) > 0$, this gives $\lambda > 1$.

Part (i)(b)

Now consider the far wall. The path is a downward parabola which at $x = d$ is above the base level $-2d$. It reaches $y = -2d$ (the base) before $x = 2d$ exactly when it is already below that level at $x = 2d$, that is when $y(2d) < -2d$. If instead $y(2d) \geq -2d$, it arrives at $x = 2d$ without having reached the base, so it touches the far wall (or clears the trench altogether).

? **Ask yourself** Why is it enough to look at the single point $x = 2d$?

Because $y(x) + 2d$ is a quadratic in x with negative leading coefficient that is positive at $x = d$. It has exactly one root beyond $x = d$, and that root lies before $2d$ precisely when the quadratic is already negative at $x = 2d$.

So we need

$$y(2d) = 2dt - \frac{4d^2(1+t^2)}{2\lambda d} < -2d \iff \lambda t - (1+t^2) < -\lambda \iff t^2 - \lambda t + 1 - \lambda > 0,$$

multiplying by $\lambda/(2d) > 0$. Multiplying by 4 and completing the square,

$$4t^2 - 4\lambda t + \lambda^2 > \lambda^2 - 4 + 4\lambda, \qquad \text{i.e.} \qquad (2\tan\alpha - \lambda)^2 > \lambda^2 + 4(\lambda - 1),$$

as required.

Deducing $\alpha > 45°$. By (a), $\lambda > 1$, so $\lambda^2 + 4(\lambda - 1) > \lambda^2 > 0$. Write $s = \sqrt{\lambda^2 + 4\lambda - 4}$, so that $s > \lambda$. The inequality in (b) says

$$2t > \lambda + s \qquad \text{or} \qquad 2t < \lambda - s.$$

The second option gives $t < 0$, because $s > \lambda$. But (a) forces $t > 0$: from $(t - \lambda)^2 < \lambda^2 - 1 < \lambda^2$ we get $|t - \lambda| < \lambda$, so $t > 0$. (Physically, the particle must rise to clear the near edge.) Hence the first option holds, and

$$2t > \lambda + s > \lambda + \lambda = 2\lambda > 2,$$

so $\tan\alpha > \lambda > 1$. Since $0 < \alpha < 90°$, this gives $\alpha > 45°$.

Part (ii)

Conversely, the two conditions together are enough. If $y(d) > 0$ then, as $y(0) = 0$ and y is concave, $y > 0$ on the whole interval $0 < x \leq d$, so the particle clears the ground and the near wall. If also $y(2d) < -2d$, it reaches the base at some x strictly between d and $2d$ and stays inside the trench until then. So we need only show that, for every $\lambda > 1$, some $t > 0$ satisfies both inequalities.

From (a), the allowed values are $\lambda - \sqrt{\lambda^2 - 1} < t < \lambda + \sqrt{\lambda^2 - 1}$. From (b), together with $t > 0$, they are $t > \frac{1}{2}(\lambda + s)$. Both hold for some t provided

$$\tfrac{1}{2}(\lambda + s) < \lambda + \sqrt{\lambda^2 - 1} \iff s < \lambda + 2\sqrt{\lambda^2 - 1}.$$

(The lower bound in (a) causes no trouble, since $\frac{1}{2}(\lambda + s) > \lambda > \lambda - \sqrt{\lambda^2 - 1}$.) Both sides are positive, so we may square:

$$\begin{aligned} & \lambda^2 + 4\lambda - 4 < \lambda^2 + 4\lambda\sqrt{\lambda^2 - 1} + 4(\lambda^2 - 1) \\ \iff \quad & 4\lambda < 4\lambda\sqrt{\lambda^2 - 1} + 4\lambda^2 \\ \iff \quad & 1 < \sqrt{\lambda^2 - 1} + \lambda, \end{aligned}$$

dividing by $4\lambda > 0$. The last line is true because $\lambda > 1$. So the interval

$$\tfrac{1}{2}\left(\lambda + \sqrt{\lambda^2 + 4\lambda - 4}\right) < \tan\alpha < \lambda + \sqrt{\lambda^2 - 1}$$

is non-empty. Choosing α with $\tan\alpha$ in it (possible, since tan takes every positive value on $(0^\circ, 90^\circ)$) makes the particle land on the base without touching either side.

Checks

- Take $\lambda = 2$. The interval is $1 + \sqrt{2} < \tan\alpha < 2 + \sqrt{3}$, which is exactly $67.5^\circ < \alpha < 75^\circ$. Trying $\alpha = 70^\circ$ in $y = xt - x^2(1 + t^2)/(4d)$ gives $y(d) \approx 0.61d > 0$ and $y(2d) \approx -3.05d < -2d$, as it should.
- As $\lambda \to 1^+$, both ends of the interval tend to 1, so the window shrinks to $\alpha = 45^\circ$: with $v^2 = dg$ the particle can only just reach the near edge, which needs the maximum-range angle.
- A computer check of thousands of random (λ, α) pairs, stepping the motion forward in time, agreed with the two inequalities in every case.

Common slips

- Treating the conditions as $y(d) > -2d$ (above the base) instead of $y(d) > 0$: the near wall rises all the way to ground level.
- Forgetting to rule out the branch $2\tan\alpha < \lambda - s$ in (b), which needs the fact that $\tan\alpha > 0$.
- Squaring an inequality in (ii) without first noting that both sides are positive.

Beyond STEP

Here a physical condition ("lands in the trench") becomes a region in the space of launch parameters, cut out by inequalities, and the question is whether that region is non-empty. This is the starting point of optimisation and control: describing the set of feasible inputs and finding its boundary. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and its extensions (air resistance, rotating frames) at first-year level.

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